

Encoding $\llbracket \cdot \rrbracket : \text{CCS-VP} \rightarrow \text{CCS}$

$$\llbracket a(x).P \rrbracket = \sum_{n \in \mathbb{N}} a_n. \llbracket P\{^n/x\} \rrbracket$$

$$\llbracket \bar{a}(e).P \rrbracket = \bar{a}_m. \llbracket P \rrbracket \quad e \downarrow m$$

$$\llbracket \tau.P \rrbracket = \tau. \llbracket P \rrbracket$$

$$\llbracket P \mid Q \rrbracket = \llbracket P \rrbracket \mid \llbracket Q \rrbracket$$

$$\llbracket \sum_{i \in I} P_i \rrbracket = \sum_{i \in I} \llbracket P_i \rrbracket$$

$$\llbracket P \setminus L \rrbracket = \llbracket P \rrbracket \setminus \{a_m \mid a \in L, m \in \mathbb{N}\}$$

$$\llbracket P[f] \rrbracket = \llbracket P \rrbracket[f'] \quad f'(a_m) = f(a)_m$$

$$\llbracket \text{if } b \text{ then } P \rrbracket = \begin{cases} \llbracket P \rrbracket & b \text{ true} \\ 0 & \text{otherwise} \end{cases}$$

$$\llbracket K(e_1, \dots, e_h) \rrbracket = K_{n_1, \dots, n_h} \quad e_i \downarrow n_i$$

$$K_{n_1, \dots, n_h} \stackrel{\text{def}}{=} \llbracket P\{^{n_1}/x_1, \dots, ^{n_h}/x_h\} \rrbracket$$

$$\text{where } K(x_1, \dots, x_h) \stackrel{\text{def}}{=} P$$

CCS Operational Semantics

$$\begin{array}{c}
\frac{}{\alpha.P \xrightarrow{\alpha} P} \text{ACT} \\
\\
\frac{P_j \xrightarrow{\alpha} P'_j}{\sum_{i \in I} P_i \xrightarrow{\alpha} P'_j} \text{SUM} \quad j \in I \\
\\
\frac{P \xrightarrow{\alpha} P'}{P \mid Q \xrightarrow{\alpha} P' \mid Q} \text{PAR-L} \\
\\
\frac{Q \xrightarrow{\alpha} Q'}{P \mid Q \xrightarrow{\alpha} P \mid Q'} \text{PAR-R} \\
\\
\frac{P \xrightarrow{\alpha} P' \quad Q \xrightarrow{\bar{\alpha}} Q'}{P \mid Q \xrightarrow{\tau} P' \mid Q'} \text{COM} \\
\\
\frac{P \xrightarrow{\alpha} P'}{P \setminus L \xrightarrow{\alpha} P' \setminus L} \text{RES} \quad \alpha, \bar{\alpha} \notin L \\
\\
\frac{P \xrightarrow{\alpha} P'}{P[f] \xrightarrow{f(\alpha)} P'[f]} \text{REN} \\
\\
\frac{P \xrightarrow{\alpha} P'}{K \xrightarrow{\alpha} P'} \text{CONST} \quad K \stackrel{\text{def}}{=} P
\end{array}$$

For all P in CCS-VP

$$\begin{aligned} \text{(i)} \quad & P \xrightarrow{\alpha} P' \implies \llbracket P \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P' \rrbracket \\ \text{(ii)} \quad & \llbracket P \rrbracket \xrightarrow{\hat{\alpha}} Q \implies \exists P'. P \xrightarrow{\alpha} P' \wedge \llbracket P' \rrbracket = Q \end{aligned}$$

$$\text{where } \hat{\alpha} = \begin{cases} a_m & \alpha = a(m) \\ \bar{a}_m & \alpha = \bar{a}(m) \\ \tau & \alpha = \tau \end{cases}$$

Both implications are established by *rule induction*, or induction on the depth of the derivation tree, which is the natural choice here because the transition relation is itself defined inductively by inference rules. The natural alternative would be *structural induction on P* , case-splitting on its syntactic form rather than on the last rule fired. For most forms this coincides with rule induction, but it fails cleanly at CONST: a constant K is a syntactic atom with no sub-term, so structural induction provides no smaller process to apply the IH to. Rule induction avoids this by measuring derivation depth, which strictly decreases at every premise, including the unfolding step inside CONST.

For (i) we proceed by induction on the derivation of $P \xrightarrow{\alpha} P'$, with a case analysis on the last rule applied. The *base cases* are the three axiom forms of CCS-VP (input, output, and τ prefix): for each, the encoding $\llbracket P \rrbracket$ is a concrete CCS term that admits the required transition directly via the Act or Sum rule. For the *inductive cases* (Sum, Par-L, Par-R, Com, Res, Ren, Const, Cond), the induction hypothesis supplies transitions for the shorter sub-derivation(s), and the matching CCS rule then closes the goal.

For (ii) we instead proceed by induction on the derivation of $\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} Q$, with a case analysis on the syntactic form of P . For each form of P , the structure of $\llbracket P \rrbracket$ uniquely determines which CCS rule could have fired last; the induction hypothesis then recovers a CCS-VP witness P' with $P \xrightarrow{\alpha} P'$, and the encoding equations confirm $\llbracket P' \rrbracket = Q$.

Throughout both directions the key invariant is that $\hat{\alpha}$ correctly bridges CCS-VP labels and CCS channels $a(m) \leftrightarrow a_m$, $\bar{a}(m) \leftrightarrow \bar{a}_m$, $\tau \leftrightarrow \tau$ ensuring that complementary actions in CCS-VP correspond to complementary channels in CCS, so that synchronization is preserved by the encoding.

$$P \xrightarrow{\alpha} P' \implies \llbracket P \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P' \rrbracket$$

Base cases

Input $a(x).P \xrightarrow{a(m)} P\{m/x\}$:

$$\frac{\frac{a_m.\llbracket P\{m/x\} \rrbracket \xrightarrow{a_m} \llbracket P\{m/x\} \rrbracket}{\sum_{n \in \mathbb{N}} a_n.\llbracket P\{n/x\} \rrbracket \xrightarrow{a_m} \llbracket P\{m/x\} \rrbracket} \text{Sum}}{\llbracket a(x).P \rrbracket \xrightarrow{a_m} \llbracket P\{m/x\} \rrbracket} =$$

Output $\bar{a}(e).P \xrightarrow{\bar{a}(m)} P, e \downarrow m$:

$$\frac{\frac{\bar{a}_m.\llbracket P \rrbracket \xrightarrow{\bar{a}_m} \llbracket P \rrbracket}{\llbracket \bar{a}(e).P \rrbracket \xrightarrow{\bar{a}_m} \llbracket P \rrbracket} \text{Act}}{=} =$$

Silent $\tau.P \xrightarrow{\tau} P$:

$$\frac{\frac{\tau.\llbracket P \rrbracket \xrightarrow{\tau} \llbracket P \rrbracket}{\llbracket \tau.P \rrbracket \xrightarrow{\tau} \llbracket P \rrbracket} \text{Act}}{=} =$$

Inductive cases

$$\frac{P_i \xrightarrow{\alpha} P'_i}{\llbracket P_i \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_i \rrbracket} \text{IH}$$

Sum $\sum_{i \in I} P_i \xrightarrow{\alpha} P'_j \quad j \in I$:

$$\frac{\frac{\frac{P_j \xrightarrow{\alpha} P'_j}{\llbracket P_j \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_j \rrbracket} \text{IH}}{\sum_{i \in I} \llbracket P_i \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_j \rrbracket} \text{Sum}}{\llbracket \sum_{i \in I} P_i \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_j \rrbracket} =$$

Par-L: $P_1 \mid P_2 \xrightarrow{\alpha} P'_1 \mid P_2$

$$\frac{\frac{\frac{P_1 \xrightarrow{\alpha} P'_1}{\llbracket P_1 \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_1 \rrbracket} \text{IH}}{\llbracket P_1 \rrbracket \mid \llbracket P_2 \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_1 \rrbracket \mid \llbracket P_2 \rrbracket} \text{Par-L}}{\llbracket P_1 \mid P_2 \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_1 \mid P_2 \rrbracket} =$$

Par-R: $P_1 \mid P_2 \xrightarrow{\alpha} P_1 \mid P'_2$

$$\frac{\frac{P_2 \xrightarrow{\alpha} P'_2}{\llbracket P_2 \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_2 \rrbracket} \text{IH}}{\frac{\llbracket P_1 \rrbracket \mid \llbracket P_2 \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P_1 \rrbracket \mid \llbracket P'_2 \rrbracket}{\llbracket P_1 \mid P_2 \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P_1 \mid P'_2 \rrbracket} =} \text{Par-R}$$

Com $P_1 \mid P_2 \xrightarrow{\tau} P'_1 \mid P'_2$:

$$\frac{\frac{P_1 \xrightarrow{a(m)} P'_1}{\llbracket P_1 \rrbracket \xrightarrow{a_m} \llbracket P'_1 \rrbracket} \text{IH} \quad \frac{P_2 \xrightarrow{\bar{a}(m)} P'_2}{\llbracket P_2 \rrbracket \xrightarrow{\bar{a}_m} \llbracket P'_2 \rrbracket} \text{IH}}{\frac{\llbracket P_1 \rrbracket \mid \llbracket P_2 \rrbracket \xrightarrow{\tau} \llbracket P'_1 \rrbracket \mid \llbracket P'_2 \rrbracket}{\llbracket P_1 \mid P_2 \rrbracket \xrightarrow{\tau} \llbracket P'_1 \mid P'_2 \rrbracket} =} \text{Com}$$

Res $P \setminus L \xrightarrow{\alpha} P' \setminus L, \alpha \notin L \cup \bar{L}$:

$$\frac{\frac{P \xrightarrow{\alpha} P'}{\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P' \rrbracket} \text{IH}}{\frac{\llbracket P \rrbracket \setminus \{a_m \mid a \in L, m \in \mathbb{N}\} \xrightarrow{\hat{\alpha}} \llbracket P' \rrbracket \setminus \{a_m \mid a \in L, m \in \mathbb{N}\}}{\llbracket P \setminus L \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P' \setminus L \rrbracket} =} \text{Res}$$

Ren $P[f] \xrightarrow{f(\alpha)} P'[f]$, where $f'(a_m) = f(a)_m$ and $\widehat{f(\alpha)} = f'(\hat{\alpha})$:

$$\frac{\frac{P \xrightarrow{\alpha} P'}{\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P' \rrbracket} \text{IH}}{\frac{\llbracket P \rrbracket[f'] \xrightarrow{f'(\hat{\alpha})} \llbracket P' \rrbracket[f']}{\llbracket P[f] \rrbracket \xrightarrow{\widehat{f(\alpha)}} \llbracket P'[f] \rrbracket} =} \text{Ren}$$

Const $K(e_1, \dots, e_h) \xrightarrow{\alpha} R, e_i \downarrow n_i$:

$$\frac{\frac{P\{^{n_1}/_{x_1}, \dots\} \xrightarrow{\alpha} R}{K_{n_1, \dots} \xrightarrow{\hat{\alpha}} \llbracket R \rrbracket} \text{IH}}{\llbracket K(e_1, \dots, e_h) \rrbracket \xrightarrow{\hat{\alpha}} \llbracket R \rrbracket} =$$

Cond if b then $P \xrightarrow{\alpha} P', b$ true:

$$\frac{\frac{P \xrightarrow{\alpha} P'}{\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P' \rrbracket} \text{IH}}{\llbracket \text{if } b \text{ then } P \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P' \rrbracket} =$$

$$\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} Q \implies \exists P'. P \xrightarrow{\alpha} P' \wedge \llbracket P' \rrbracket = Q$$

Base cases

Input $[a(x).P]$

$$\frac{}{a(x).P \xrightarrow{a(m)} P\{^m/_x\}} \text{Act}$$

$$\llbracket a(x).P \rrbracket = \sum_{n \in \mathbb{N}} a_n \cdot \llbracket P\{^n/_x\} \rrbracket.$$

The only possible transition fires branch a_m for some $m \in \mathbb{N}$, so $Q = \llbracket P\{^m/_x\} \rrbracket$.

Taking $P' = P\{^m/_x\}$ witnesses $P \xrightarrow{a(m)} P'$ and $\llbracket P' \rrbracket = Q$.

Output $[\bar{a}(e).P]$

$$\frac{e \downarrow m}{\bar{a}(e).P \xrightarrow{\bar{a}(m)} P} \text{Act}$$

$$\llbracket \bar{a}(e).P \rrbracket = \bar{a}_m \cdot \llbracket P \rrbracket \text{ since } e \downarrow m.$$

The unique transition via Act forces $Q = \llbracket P \rrbracket$.

Taking $P' = P$ witnesses $P \xrightarrow{\bar{a}(m)} P'$ and $\llbracket P' \rrbracket = Q$.

Silent $[\tau.P]$

$$\frac{}{\tau.P \xrightarrow{\tau} P} \text{Act}$$

$$\llbracket \tau.P \rrbracket = \tau \cdot \llbracket P \rrbracket.$$

The unique transition via Act forces $Q = \llbracket P \rrbracket$.

Taking $P' = P$ witnesses $P \xrightarrow{\tau} P'$ and $\llbracket P' \rrbracket = Q$.

Inductive cases

$$\frac{\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} Q}{P \xrightarrow{\alpha} P' \wedge \llbracket P' \rrbracket = Q} \text{IH}$$

$$\text{Sum} \left[\sum_{i \in I} P_i \right]$$

$$\frac{\frac{\llbracket P_j \rrbracket \xrightarrow{\hat{\alpha}} Q}{P_j \xrightarrow{\alpha} P'_j \wedge \llbracket P'_j \rrbracket = Q} \text{IH}}{\sum_{i \in I} P_i \xrightarrow{\alpha} P'_j} \text{Sum}$$

Since $\llbracket \sum_{i \in I} P_i \rrbracket = \sum_{i \in I} \llbracket P_i \rrbracket$, any transition must originate from some branch $j \in I$.

The IH recovers $P_j \xrightarrow{\alpha} P'_j$ with $\llbracket P'_j \rrbracket = Q$.

Taking $P' = P'_j$ witnesses $P \xrightarrow{\alpha} P'$ and $\llbracket P' \rrbracket = Q$.

Par $[P_1 \mid P_2]$

$$\llbracket P_1 \mid P_2 \rrbracket = \llbracket P_1 \rrbracket \mid \llbracket P_2 \rrbracket.$$

Any transition of $\llbracket P_1 \rrbracket \mid \llbracket P_2 \rrbracket$ must have arisen from one of three CCS rules.

- PAR-L

$$\frac{\frac{\llbracket P_1 \rrbracket \xrightarrow{\hat{\alpha}} Q_1}{P_1 \xrightarrow{\alpha} P'_1 \wedge \llbracket P'_1 \rrbracket = Q_1} \text{IH}}{P_1 \mid P_2 \xrightarrow{\alpha} P'_1 \mid P_2} \text{Par-L}$$

The transition on the left component gives $Q = Q_1 \mid \llbracket P_2 \rrbracket$.

The IH recovers $P_1 \xrightarrow{\alpha} P'_1$ with $\llbracket P'_1 \rrbracket = Q_1$.

Taking $P' = P'_1 \mid P_2$ witnesses $P \xrightarrow{\alpha} P'$ and $\llbracket P' \rrbracket = \llbracket P'_1 \rrbracket \mid \llbracket P_2 \rrbracket = Q_1 \mid \llbracket P_2 \rrbracket = Q$.

- PAR-R

$$\frac{\frac{\llbracket P_2 \rrbracket \xrightarrow{\hat{\alpha}} Q_2}{P_2 \xrightarrow{\alpha} P'_2 \wedge \llbracket P'_2 \rrbracket = Q_2} \text{IH}}{P_1 \mid P_2 \xrightarrow{\alpha} P_1 \mid P'_2} \text{Par-R}$$

The transition on the right component gives $Q = \llbracket P_1 \rrbracket \mid Q_2$.

The IH recovers $P_2 \xrightarrow{\alpha} P'_2$ with $\llbracket P'_2 \rrbracket = Q_2$.

Taking $P' = P_1 \mid P'_2$ witnesses $P \xrightarrow{\alpha} P'$ and $\llbracket P' \rrbracket = \llbracket P_1 \rrbracket \mid \llbracket P'_2 \rrbracket = \llbracket P_1 \rrbracket \mid Q_2 = Q$.

- COM

$$\frac{\frac{\llbracket P_1 \rrbracket \xrightarrow{a_m} Q_1}{P_1 \xrightarrow{a(m)} P'_1 \wedge \llbracket P'_1 \rrbracket = Q_1} \text{IH} \quad \frac{\llbracket P_2 \rrbracket \xrightarrow{\bar{a}_m} Q_2}{P_2 \xrightarrow{\bar{a}(m)} P'_2 \wedge \llbracket P'_2 \rrbracket = Q_2} \text{IH}}{P_1 \mid P_2 \xrightarrow{\tau} P'_1 \mid P'_2} \text{Com}$$

Complementary transitions a_m and $\bar{a}_m$ map back to $a(m)$ and $\bar{a}(m)$.

The IH on each side recovers $P_1 \xrightarrow{a(m)} P'_1$ with $\llbracket P'_1 \rrbracket = Q_1$ and $P_2 \xrightarrow{\bar{a}(m)} P'_2$ with $\llbracket P'_2 \rrbracket = Q_2$.

Taking $P' = P'_1 \mid P'_2$ witnesses $P \xrightarrow{\tau} P'$ and $\llbracket P' \rrbracket = Q_1 \mid Q_2 = Q$.

Res $[P \setminus L]$

$$\frac{\frac{\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} Q_0}{P \xrightarrow{\alpha} P' \wedge \llbracket P' \rrbracket = Q_0} \text{IH}}{P \setminus L \xrightarrow{\alpha} P' \setminus L} \text{Res } (\alpha \notin L \cup \bar{L})$$

$\llbracket P \setminus L \rrbracket = \llbracket P \rrbracket \setminus \{a_m \mid a \in L, m \in \mathbb{N}\}$, so the transition decomposes as $\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} Q_0$ with $Q = Q_0 \setminus \{a_m \mid a \in L, m \in \mathbb{N}\}$.

The transition being unblocked forces $\hat{\alpha} \notin \{a_m \mid a \in L\}$, hence $\alpha \notin L \cup \bar{L}$.

The IH recovers $P \xrightarrow{\alpha} P'$ with $\llbracket P' \rrbracket = Q_0$.

Taking $P' \setminus L$ as the witness gives $P \setminus L \xrightarrow{\alpha} P' \setminus L$ and $\llbracket P' \setminus L \rrbracket = \llbracket P' \rrbracket \setminus \{a_m \mid a \in L\} = Q_0 \setminus \{a_m \mid a \in L\} = Q$.

Ren [$P[f]$]

$$\frac{\frac{\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} Q_0}{P \xrightarrow{\alpha} P' \wedge \llbracket P' \rrbracket = Q_0} \text{IH}}{P[f] \xrightarrow{f(\alpha)} P'[f]} \text{Ren}$$

$\llbracket P[f] \rrbracket = \llbracket P \rrbracket[f']$, so the transition decomposes as $\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} Q_0$ with $Q = Q_0[f']$.

The IH recovers $P \xrightarrow{\alpha} P'$ with $\llbracket P' \rrbracket = Q_0$.

Taking $P'[f]$ as the witness gives $P[f] \xrightarrow{f(\alpha)} P'[f]$ and $\llbracket P'[f] \rrbracket = \llbracket P' \rrbracket[f'] = Q_0[f'] = Q$.

Cond [if b then P]

$$\frac{b \text{ true} \quad \frac{\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} Q}{P \xrightarrow{\alpha} P' \wedge \llbracket P' \rrbracket = Q} \text{IH}}{\text{if } b \text{ then } P \xrightarrow{\alpha} P'} \text{Cond}$$

$\llbracket \text{if } b \text{ then } P \rrbracket = \llbracket P \rrbracket$ when b is true.

The transition is directly on $\llbracket P \rrbracket$.

The IH recovers $P \xrightarrow{\alpha} P'$ with $\llbracket P' \rrbracket = Q$.

The IH witness P' directly satisfies $P \xrightarrow{\alpha} P'$ and $\llbracket P' \rrbracket = Q$.

Const [$K(e_1, \dots, e_h)$]

$$\frac{e_1 \downarrow n_1, \dots, e_h \downarrow n_h \quad \frac{\llbracket P\{^{n_1}/_{x_1}, \dots\} \rrbracket \xrightarrow{\hat{\alpha}} Q}{P\{^{n_1}/_{x_1}, \dots\} \xrightarrow{\alpha} R \wedge \llbracket R \rrbracket = Q} \text{IH}}{K(e_1, \dots, e_h) \xrightarrow{\alpha} R} \text{Const}$$

$\llbracket K(e_1, \dots, e_h) \rrbracket = K_{n_1, \dots} \stackrel{\text{def}}{=} \llbracket P\{^{n_i}/_{x_i}\} \rrbracket$.

The transition is on that encoding.

The IH recovers $P\{^{n_1}/_{x_1}, \dots\} \xrightarrow{\alpha} R$ with $\llbracket R \rrbracket = Q$.

Taking $P' = R$ witnesses $K(e_1, \dots, e_h) \xrightarrow{\alpha} P'$ and $\llbracket P' \rrbracket = Q$.

The interpreter developed for this project encodes input actions over a bounded interval $[n_{\min}, n_{\max}] \subset \mathbb{N}$ rather than an infinite sum over all of $\mathbb{N}$. Concretely, $\llbracket a(x).P \rrbracket = \sum_{n=n_{\min}}^{n_{\max}} a_n \cdot \llbracket P\{^n/x\} \rrbracket$ instead of $\sum_{n \in \mathbb{N}} a_n \cdot \llbracket P\{^n/x\} \rrbracket$. The proof above remains structurally identical: the **Input** base case still applies the Sum rule and the $=$ step, and every inductive case goes through unchanged. The only difference is that operational correspondence holds under the assumption that all communicated values fall within the chosen interval; values outside it have no corresponding branch in the encoding and therefore no matching CCS transition.